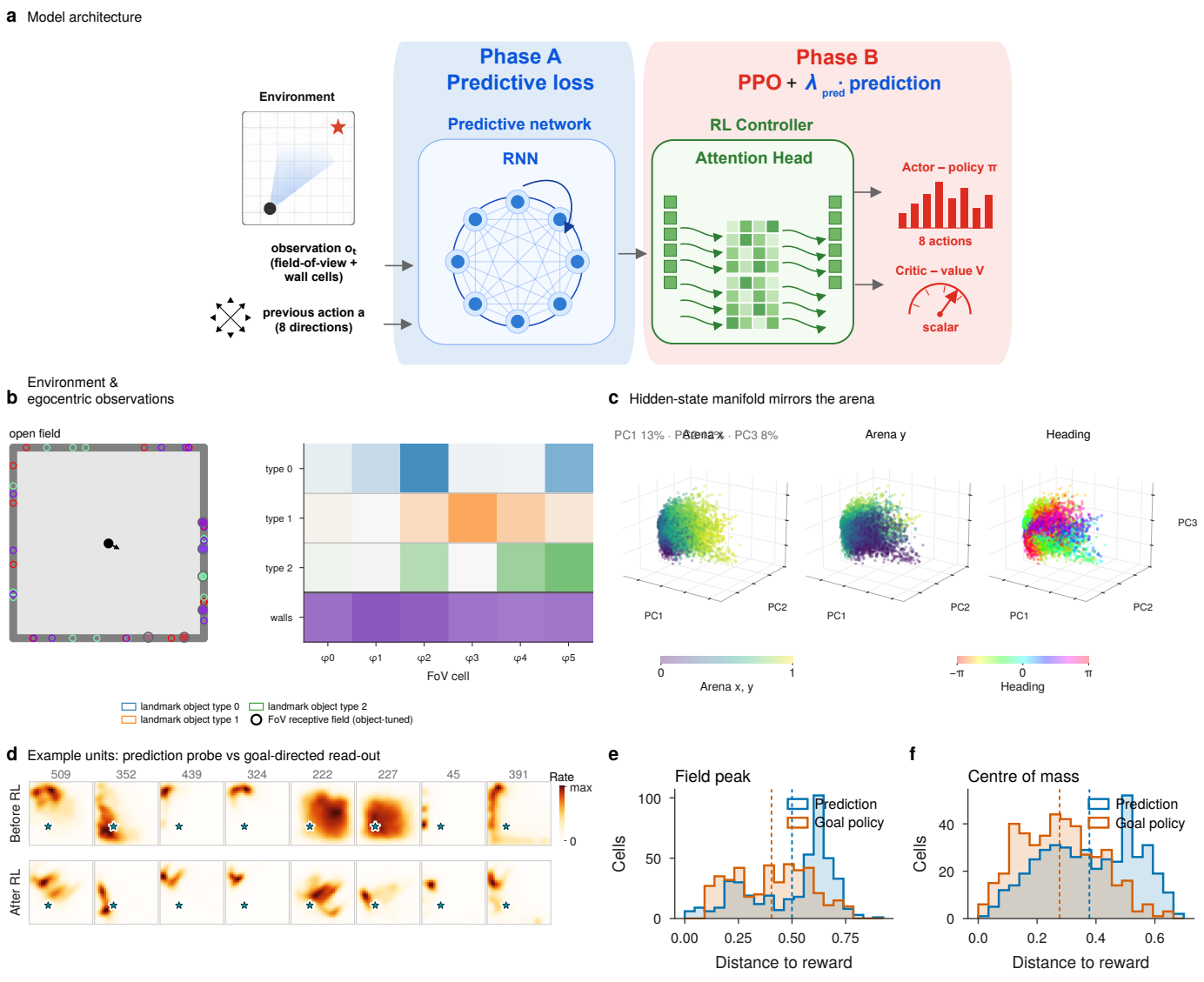

a Model architecture
Environment
observation $o_t$ (field-of-view + wall cells)
previous action a (8 directions)
Phase A
Predictive loss
Predictive network
RNN
Phase B
PPO + $\lambda_{pred}$ prediction
RL Controller
Attention Head
Actor – policy π
8 actions
Critic – value V
scalar
b Environment & egocentric observations
open field
type 0
type 1
type 2
walls
φ0
φ1
φ2
φ3
φ4
φ5
FoV cell
landmark object type 0
landmark object type 1
landmark object type 2
FoV receptive field (object-tuned)
c Hidden-state manifold mirrors the arena
PC1 13% · PC2 9% · PC3 8%
Arena x
Arena y
Heading
PC1
PC2
PC3
0
1
Arena x, y
−π
0
π
Heading
d Example units: prediction probe vs goal-directed read-out
509
352
439
324
222
227
45
391
Rate
max
0
Before RL
After RL
e
Field peak
Prediction
Goal policy
Cells
100
50
0
0.00
0.25
0.50
0.75
Distance to reward
f
Centre of mass
Prediction
Goal policy
Cells
40
20
0
0.0
0.2
0.4
0.6
Distance to reward